

■ 1.3.9 Special Topic: Using Files

If you use *Mathematica* with a graphical front end, you will probably never need to know about the file manipulation capabilities of *Mathematica* discussed in this section.

You can store *Mathematica* input and output in standard (plain text) files, which you can, for example, read and edit with a text editor.

When you are building up large calculations, you will often find it convenient to store some of your intermediate results in files. You can also store *Mathematica* definitions in files, and read them in when you need them. In fact, when you are making complicated definitions, you may find it best to work on them in a file, using an external text editor, and only later read them into *Mathematica*. If you subsequently want to change the definitions, you can just edit the file, and read it in again. (If you want to work like this, you should make sure it does not cause any trouble to execute the commands in your files more than once.)

It is a convention that all files containing *Mathematica* input have names that end with a `.m` suffix. When you look through all the files you have, you can then quickly recognize which of them are *Mathematica* input.

<code><<name</code>	read in a file of <i>Mathematica</i> input
<code>!!name</code>	display the contents of a file
<code>Save["name", x₁, x₂, ...]</code>	save definitions for variables
<code>x >> name</code>	put the value of <i>x</i> into the file <i>name</i>
<code>x >>> name</code>	append the value of <i>x</i> to <i>name</i>
<code>!command</code>	execute an external (operating system) command (available only on some computer systems)

Functions for manipulating files without using a graphical front end.

This sets the variable *x* to 6.

```
In[1]:= x = 6
Out[1]= 6
```

This saves the definition you have made for *x* in the file `tmp.m`. Do not forget to put quotes around the file name.

```
In[2]:= Save["tmp.m", x]
```

Web sample page from The *Mathematica* Book, First Edition, by Stephen Wolfram, published by Addison-Wesley Publishing Company (hardcover ISBN 0-201-19334-5; softcover ISBN 0-201-19330-2). To order *Mathematica* or this book contact Wolfram Research: info@wolfram.com; <http://www.wolfram.com/>; 1-800-441-6284.

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This displays the contents of the file `tmp.m`.
You do not need quotes around the file
name here.

```
In[3]:= !!tmp.m
x = 6
```

This sets `y` to 7.

```
In[3]:= y = 7
Out[3]= 7
```

This adds the definition of `y` to the file
`tmp.m`.

```
In[4]:= Save["tmp.m", y]
```

The file `tmp.m` now contains definitions for
both `x` and `y`.

```
In[5]:= !!tmp.m
x = 6
y = 7
```

This reads in the file `tmp.m`, executing the
assignments for the variables `x` and `y` just
as if you had typed them directly into
Mathematica.

```
In[5]:= <<tmp.m
Out[5]= 7
```

This executes the external operating
system command `rm tmp.m`. What external
commands, if any, you can execute, will
depend on what particular computer
system you are using. This example is for a
standard UNIX operating system. The
command removes the file `tmp.m`.

```
In[6]:= !rm tmp.m
```

This calculates the value of `10!`, and puts
the result in the file `tmp`.

```
In[6]:= 10! >> tmp
```

Now the file `tmp` contains the value of `10!`.

```
In[7]:= !!tmp
3628800
```

Reading in `tmp` gives the result you stored.

```
In[7]:= <<tmp
Out[7]= 3628800
```

This *appends* `20!` to the file `tmp`. If you use
`>>` instead of `>>>`, you overwrite what was
previously in the file.

```
In[8]:= 20! >>> tmp
```

If you are familiar with the UNIX shell, you will recognize the *Mathematica* operators `>>` and `<<` as being analogous to the UNIX redirection operators `>` and `<`. The *Mathematica* operator `>>>` is analogous to the UNIX operator `>>`. (*Mathematica* uses the single characters `>` and `<` to represent the standard relational operators “greater than” and “less than”.)

When you ask for a file using `<<`, *Mathematica* will search a sequence of direc-

tories to try and find it. The directories it searches are specified by the value of `$Path`, as discussed in Section 2.6.17.